

PUBLICATION PROPOSAL

Dash3R

Semi-supervised adaptation of 3D foundation models
to in-the-wild driving video

“Train where labels exist. Adapt where they don’t. Evaluate everywhere.”



Supervised

Rig3R-V2 on labelled rigs



Self-supervised

LA-Pose on raw dashcam



GT-free eval

Reprojection benchmark

3D foundation models break on the driving long tail



Feed-forward 3D works...

VGGT-class models deliver cameras, depth and point maps in one pass — trained largely on curated datasets with 3D ground truth.



...until the domain shifts

Dashcams: unknown intrinsics, rolling shutter, dynamic traffic, night, rain, new countries. Quality drops exactly where driving needs it most.



And labels can't follow

LiDAR + RTK ground truth exists for a few rigs in a few cities. The long tail of cameras and geographies has none.

UNLIMITED

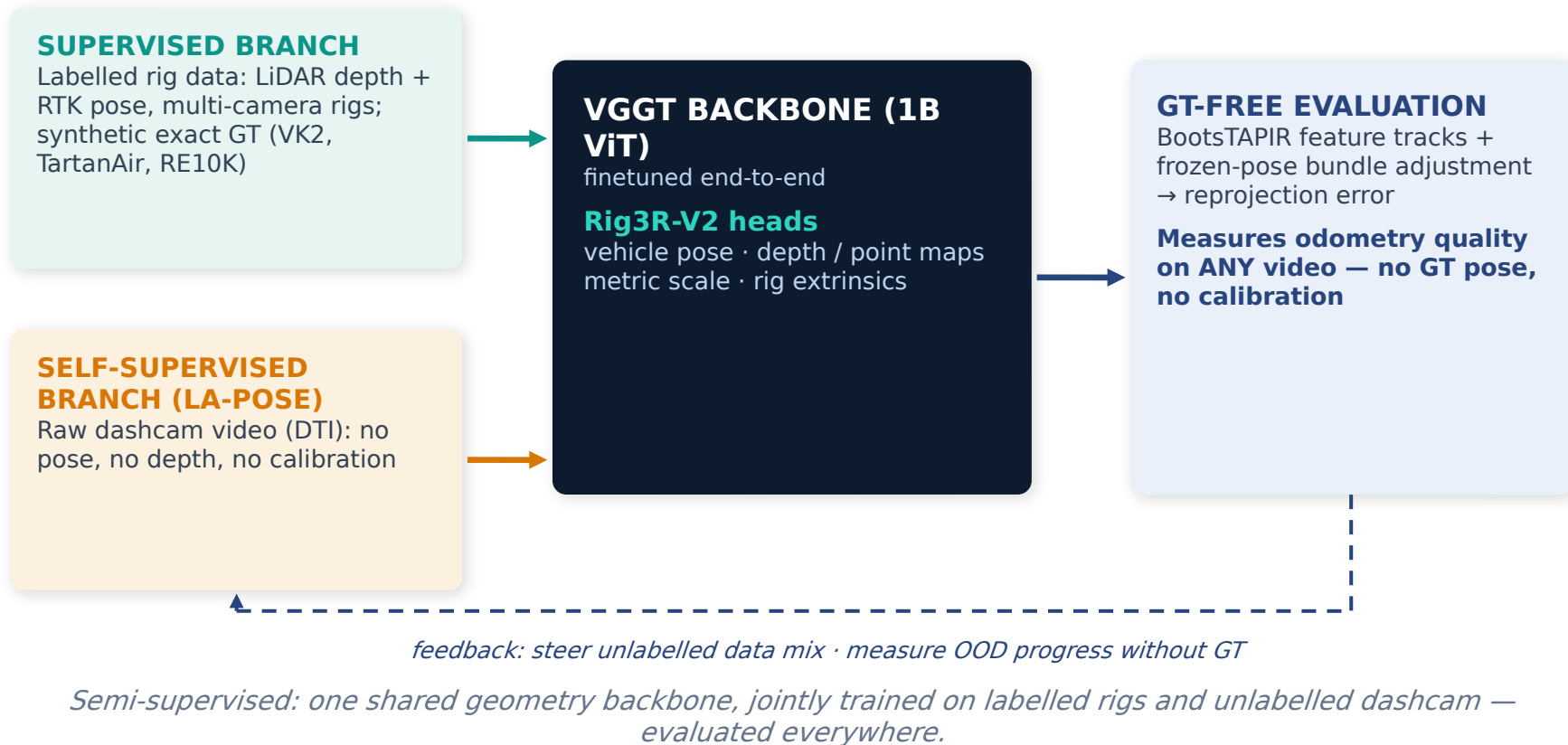
raw dashcam video — label-free,
every country, camera and weather

A HANDFUL

of rigs and cities with LiDAR + RTK
ground-truth pose and depth

*The data we most need for
generalization is the data we can
neither label nor — until now —
evaluate on.*

One recipe: two training branches, one label-free feedback loop



Branch 1 · Rig3R-V2 — VGGT finetuned on labelled driving data



Backbone

VGGT 1B ViT finetuned end-to-end (DINOv3 encoder variant for ablation).



Rig-aware, multi-camera

Rig3R lineage: 5-6 camera rigs, conditioning on and prediction of rig extrinsics.



Heads

Vehicle SE(3) pose · depth or point maps · metric scale · camera extrinsics.



Supervision

LiDAR depth + RTK trajectories from real fleets (Wayve Gen1/Gen2 + partner data); synthetic exact-GT (Virtual KITTI 2, TartanAir V2, RE10K).

What labels buy

- Metric scale grounded by RTK
- Calibrated multi-camera geometry
- Clean, dense depth supervision
- Stable trajectories at highway speed

Strong in-distribution — but every labelled hour comes from a rig we own. The open world doesn't.

Branch 2 · LA-Pose — self-supervised ego-motion from raw dashcam



Input

Unlabelled monocular dashcam clips (e.g. DTI): no pose, no depth, calibration unknown and different in every run.



Model

Latent-geometry inverse dynamics: image features + cross-frame transformer predict inter-frame translation and rotation.



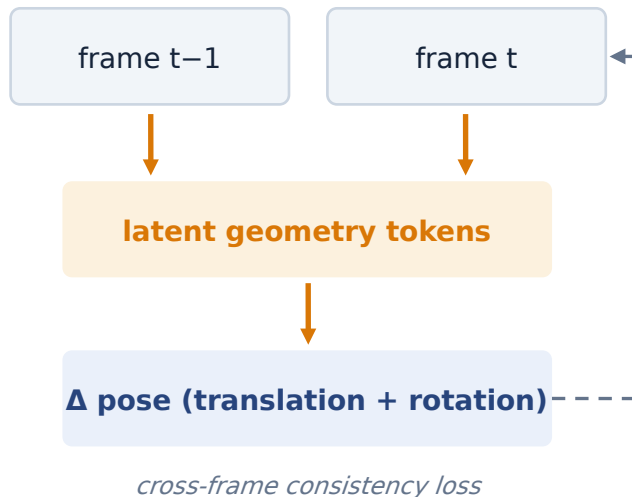
Objective

Self-supervision from cross-frame visual consistency — no human labels anywhere in the loop.



Scale

Weak ego signals (e.g. wheel speed) can anchor metric scale where available.



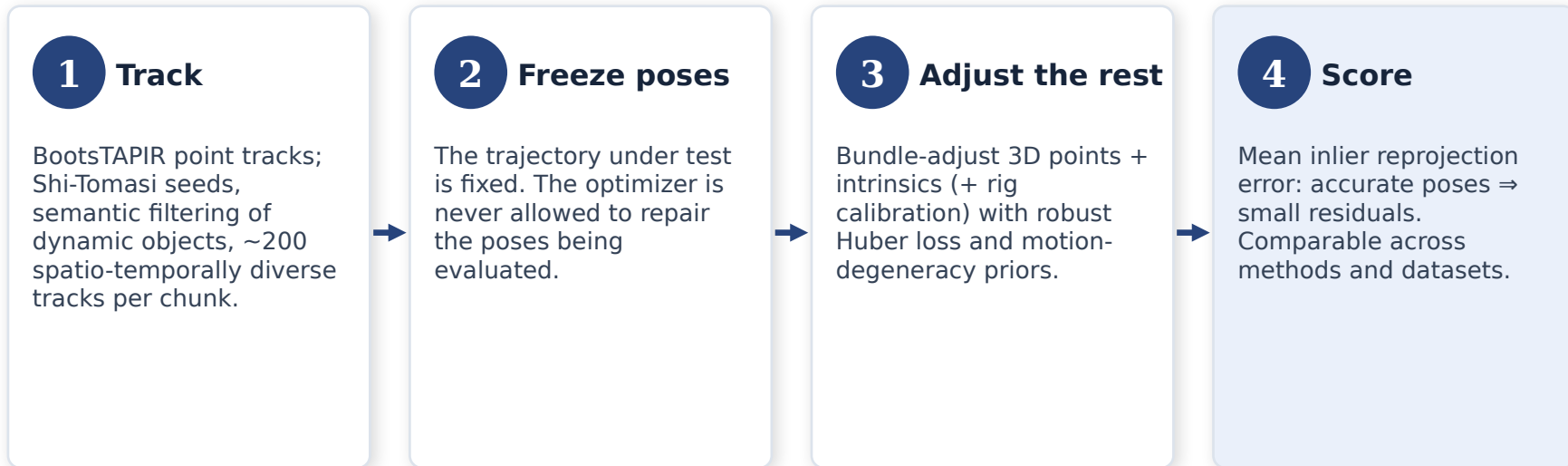
This branch lets the backbone see the target domain: every camera, country and weather the labelled fleet never visits.

Why joint training: labels give metric truth, video gives the distribution

Capability	Supervised only	Self-supervised only	Semi-supervised (ours)
Metric scale & calibration	✓	✗	✓
Multi-camera rig geometry	✓	✗	✓
Adapts to unlabelled dashcam	✗	✓	✓
New geographies, cameras, weather	✗	✓	✓
Dense geometry (depth / points)	✓	~	✓

Precedent: Noisy Student and FixMatch showed unlabelled data lifts recognition at scale. Nobody has demonstrated the same for feed-forward 3D geometry on a foundation model — **that demonstration is the paper.**

GT-free evaluation — frozen-pose reprojection error



Runs on any video — no GT pose, no calibration. Already deployed over 1,000 $\times$ 1-min DTI Japan segments and the Gen2 RTK benchmarks.

Internal: “GT-Free Odometry Evaluation” (Notion) · [spatial/eval/feature_tracking benchmark registry](#)

“You train and test on reprojection” — pre-empting the reviewer

Why it isn't circular



External signal

Evaluation uses an independent SOTA tracker (BootsTAPIR) — not the model's own features, losses or correspondences.



Frozen poses

Bundle adjustment can never repair the trajectory under test; only points and intrinsics move.



Degeneracy handled

Priors keep stopped / low-parallax motion well-posed without favouring any method.

Validation we must ship



Correlate with GT

Rank-correlate the GT-free score with ATE/RPE wherever GT exists (Gen2 RTK, KITTI) — the headline validation figure.



Noise injection

Perturb poses with controlled error; the metric must respond monotonically.



Tracker independence

Replicate with a second tracker (e.g. CoTracker); report GT metrics wherever GT exists, GT-free everywhere else.

What we already see internally

DASHCAM (DTI)

SSL wins

LA-Pose trained on raw dashcam outperforms the VGGT baseline on the GT-free reprojection benchmark.

numbers: odometry dashboard

LABELLED RIGS

Parity+

Rig3R-V2 is competitive with the classical odometry stack on Gen2 RTK benchmarks.

numbers: Rig3R-V2 ablation page

EVAL AT SCALE

1,000 segs

GT-free pipeline already runs over 1,000 × 1-min DTI Japan segments; >90% of tracks pass visual inspection.

source: GT-free eval doc

Exact numbers intentionally omitted here — pull from the odometry dashboard and the Rig3R-V2 ablation pages before circulating this deck.

Experiments the paper needs

Block	What we run	Why it matters
1 • Validate the metric	GT-free $\leftrightarrow$ ATE/RPE rank correlation on Gen2 RTK + KITTI; pose-noise sensitivity sweep; cross-tracker check	<i>Makes the benchmark a citable contribution</i>
2 • Main OOD results	Public GT: KITTI, nuScenes, Waymo, DDAD. GT-free on public dashcam (OpenDV-2K / BDD100K) + internal DTI	<i>Reproducible story beyond Wayve data</i>
3 • Baselines	Feed-forward: VGGT, MAST3R, Pi3, MapAnything, CUT3R. Classical: COLMAP, DROID-SLAM, DPVO	<i>Positions against both worlds</i>
4 • Ablations	Supervised-only vs SSL-only vs semi-supervised at matched compute; scaling in unlabelled hours; forgetting on generic 3D benchmarks	<i>Headline figure: OOD gap vs unlabelled data</i>

The scaling curve — OOD error vs unlabelled hours — is the figure reviewers will remember.

The claims

1

Recipe

Dash3R: first semi-supervised adaptation of a 3D foundation model to unlabelled driving video at scale.

2

Benchmark

A GT-free odometry evaluation protocol — toolkit and public dashcam splits released.

3

Results

State-of-the-art OOD ego-motion on dashcam domains, with no in-distribution regression.

4

Analysis

Scaling study: how far do unlabelled hours close the OOD gap of 3D foundation models?

Net effect: any fleet, any camera, any country — we can improve the model there and measure it there, without ground truth.

Risks & open questions



Circularity optics — Reviewers may distrust reprojection as both training signal and score — ship the validation block (slide 8) before anything else.



Proprietary data — Internal corpora and benchmarks can't be released — public-data track (OpenDV-2K / BDD100K) plus toolkit release; needs approval.



Naming — “DAS3R” (2024, dynamic scene splatting) is uncomfortably close to “Dash3R” — verify clearance; “Rig3R-V2” remains the fallback.

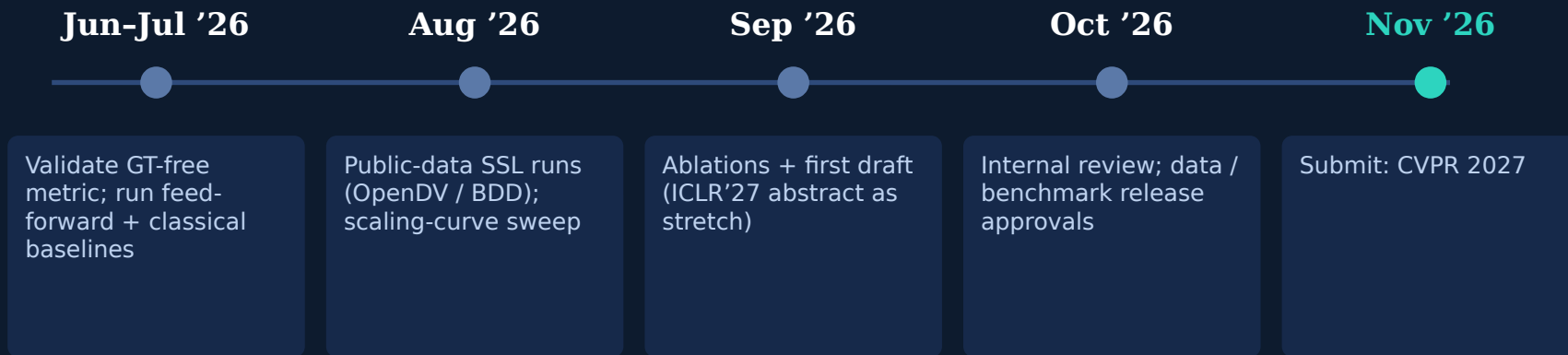


Dynamic scenes — Moving objects can pollute evaluation tracks; stronger per-frame semantic filtering is a known TODO of the eval pipeline.



Attribution — Is the OOD gain from self-supervision, or simply from more data? Matched-compute ablations must answer this cleanly.

Path to submission



Primary: CVPR 2027 (Nov '26) · Stretch: ICLR 2027 (Sep '26) · Backup: 3DV / ICCV 2027

Dash3R: turn every unlabelled mile into training signal — and into a benchmark.